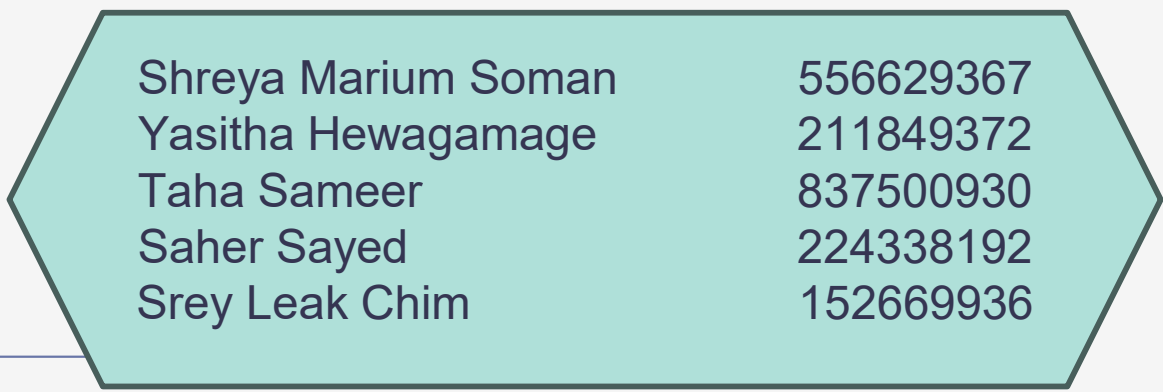


Customer Retention and Engagement Levels

Bank of Twizel

Group 13



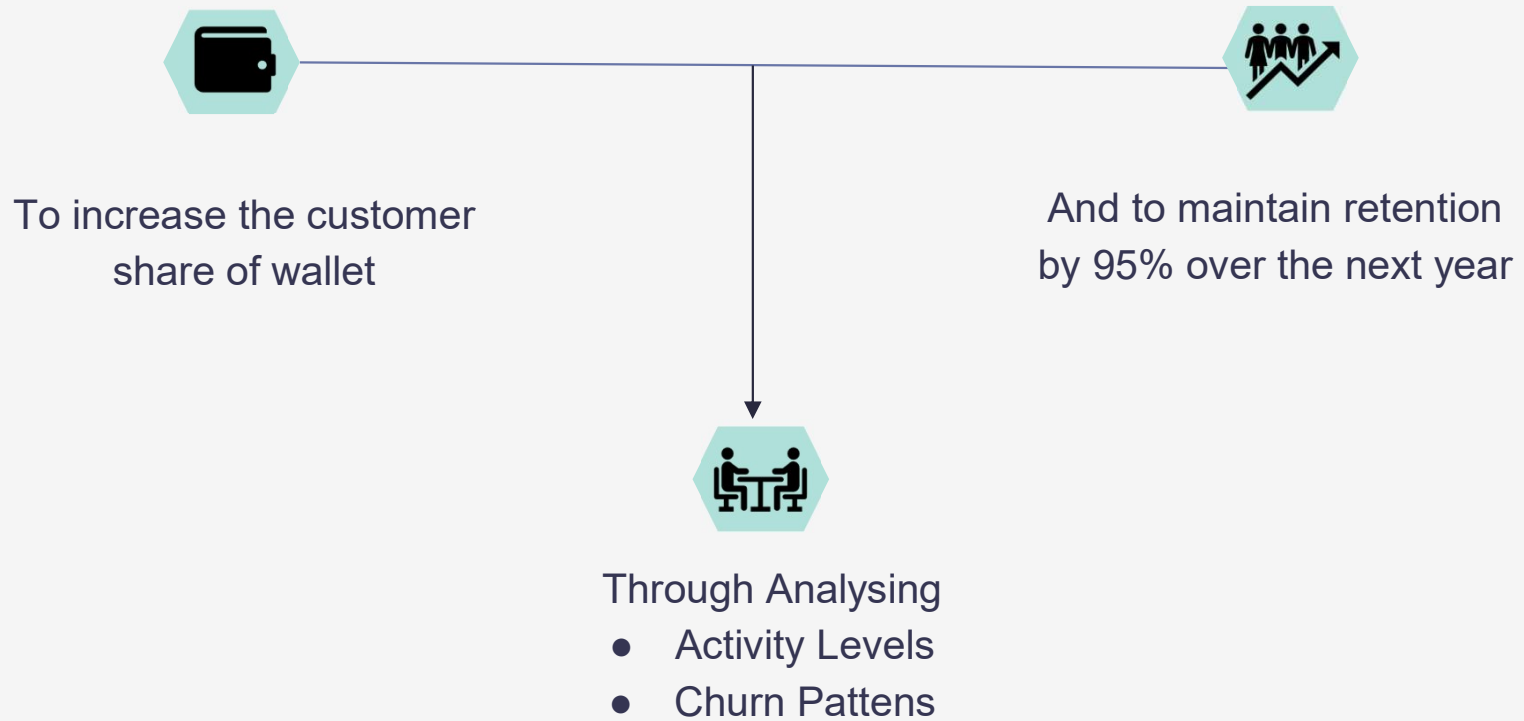
Shreya Marium Soman	556629367
Yasitha Hewagamage	211849372
Taha Sameer	837500930
Saher Sayed	224338192
Srey Leak Chim	152669936



Table of Contents

- 01** Objective
- 02** Methodology
- 03** Analysis of Client Activity Level
- 04** Analysis of Client Churn
- 05** Recommendations

Business Objective



Methodology

Approaches

```
graph TD; A[Approaches] --> B[Linear Regression]; A --> C[Logistics Regression]; B --> D[Used to analyze on the activity level of the client]; C --> E[Used to analyze the probabilities of client churn];
```

Linear Regression

Used to analyze on the
activity level of the client

Logistics Regression

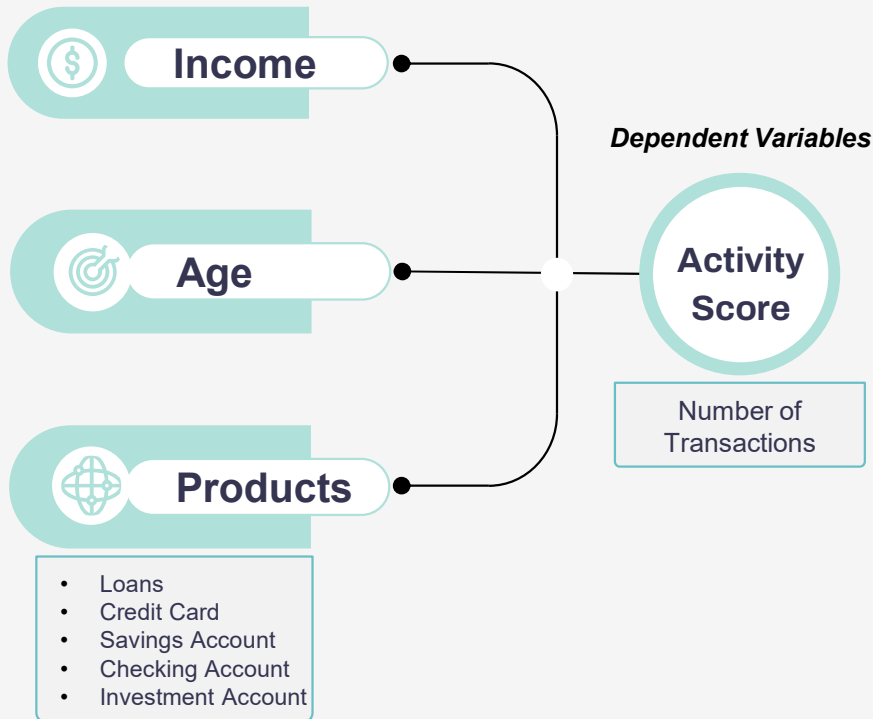
Used to analyze the
probabilities of client churn

Supervised Machine Learning Techniques

Analysis of Activity Levels

Activity Levels: $0.224 + 0.166(L) + 0.163(CC) + 1.033(CA) + 0.049(IT) + 0.234(SA) + 0.396(IgInc) - 0.160(IgAge)$

Independent Variable



Coefficient Table

Variable	Unstandardized B	Standardized Coefficient B	Sig.
Constant	.224	-	.081
Loans (L)	.166	.043	<.001
Credit Card (CC)	.163	.062	<.001
Checking A/C (CA)	1.03	.025	<.001
Investment Trust (IT)	.049	.012	<.001
Savings A/C (SA)	.234	.122	<.001
Log Income (IgInc)	.396	.679	<.001
Log Age (IgAge)	-.160	-.034	<.001

Analysis of Activity Levels

Activity Levels: $0.224 + 0.166(L) + 0.163(CC) + 1.033(CA) + 0.049(IT) + 0.234(SA) + 0.396(\lg Inc) - 0.160(\lg Age)$

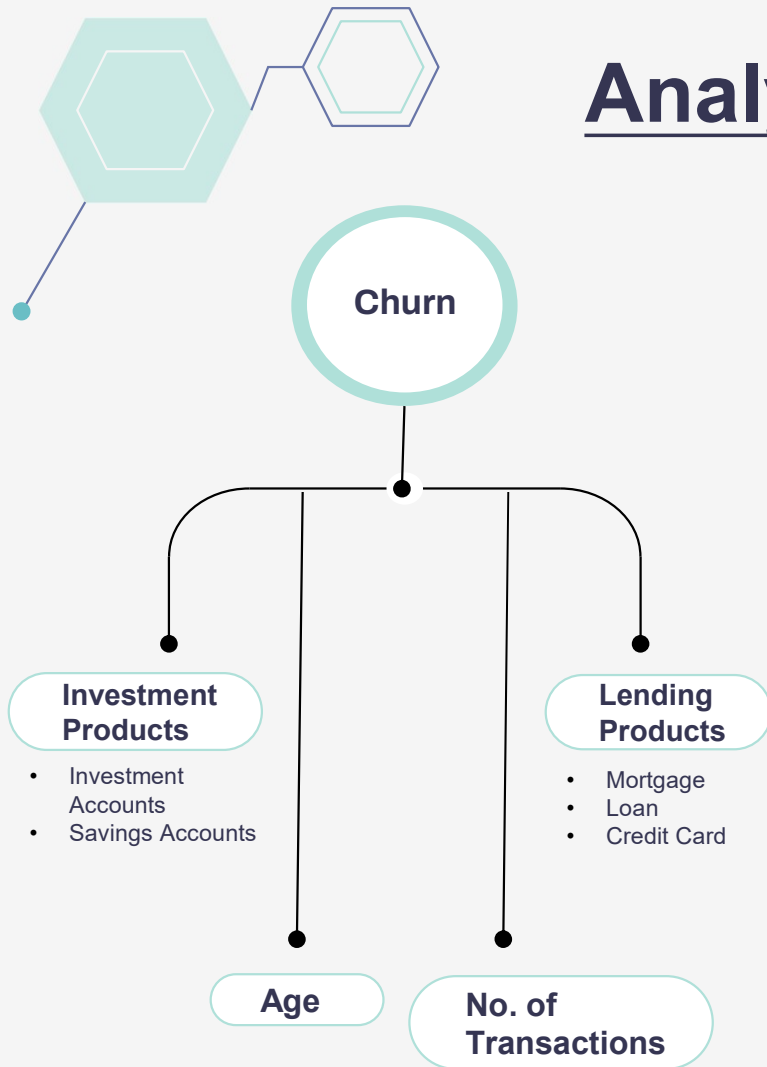
Client Profile



Name: Emily
Age: 64
Income: \$ 5,623

Variables	Weight	Score	Weight*Score
Constant	0.224		
Loans(L)	0.166	0	0
Credit Card (CC)	0.163	1	0.16
Checking A/C (CA)	1.037	1	1.03
Investment Trust (IT)	0.049	0	0
Savings A/C (SA)	0.2.34	0	0
Log Income (lgInc)	0.396	3.75 (\$5,623)	1.49
Log Age (lgAge)	-0.160	1.81 (64)	-0.29
Activity Level	2.62		
			413

Analysis of Churn



Variables in the Equation	Beta Coefficients	Exp (Beta Coefficients)	Significance
Constant	-3.759	0.023	<0.001
Loan	-0.548	0.578	0.034
Credit Card	-0.380	0.684	0.022
Investment Trust	-0.847	0.429	0.004
Mortgage	-0.648	0.523	0.043
Savings Account	-0.469	0.625	<0.001
Age	0.010	1.010	<0.001
No. of Transactions	-0.003	0.997	<0.001

Nagelkerke $R^2 = 0.032$.

Cutoff value = 0.026

$$P(\text{Churn}) = 1 / (1 + e^{-(\text{Total Score})})$$



Analysis of Churn

Variables	Weight	Score	Weight * Score
Constant	-3.759		-3.759
Loan	-0.548	0	0
Credit Card	-0.380	1	-0.380
Investment Trust	-0.847	0	0
Mortgage	-0.648	0	0
Savings Account	-0.469	1	-0.469
Age	0.010	65	0.650
No. of Transactions	-0.003	211	-0.633
Total Score			-4.591

$$P(\text{Churn}) = 1 / (1 + e^{-(-4.591)}) = 1.00\%$$

Client Profile



Name: George

Age: 65

As per our model, this customer will not be predicted to churn as $1.00\% < 2.6\%$, thus we'll not target them.

Classification Table for Churn

		Predicted		Percentage Correct
		Stay	Churn	
Observed	Stay	33258	13563	71.0
	Churn	413	497	54.6
Overall Percentage				70.7

Cutoff value = 0.026



		Predicted		Percentage Correct
		Stay	Churn	
Observed	Stay	33258	13563	71.0
	Churn	413	497	54.6
Overall Percentage				70.7

Cutoff value = 0.026

Actual Churn
1% of Total Customers

Estimates as per Kiwi Bank (Financial Statements, 2023)

Marketing Cost per customer per year

3

Revenue Earned for Retained Customer

132

Cost of Acquiring a New Customer

15



Target group

14,060 (29%)

736,419

\$2.20 million
Marketing

Customers not targeted

413 (0.87%)

21,683

\$2.85 million
Opportunity Loss

Predicted to stay

13,563 (96.5%)

710,387

\$93.77 million
Revenue

Net Profit: \$ 88.32 million

Key Strategies



OWNING A MORTGAGE REDUCES CHURN

Incentivize Mortgage

- If the mortgage period is fixed for at least 3 years (or decided by the management)
- The customer will be entitled to a special savings account.

Saving Accounts Interest Rate

- Receive exclusive Interest Rate for Saving Accounts during the term of the mortgage
- Savings account be used to pay the instalments of the mortgage with fee waivers
- Campaign this bundle offer though risk management

Exclusive Premium Product Bundling

- Maintains a minimum balance to access other premium products like an **Elite Credit Card** with low mark-up and lower annual fee.



Key Strategies

Strategies



Income Transfer Strategy

Savings Accounts

- Incentivize direct transfer of salary to a special savings account
- Two transaction per month any other account
- Enjoy premium interest rates based on a minimum account balance decided by the management.

Credit Cards

- Cross-selling credit card through payroll account set up (limit is based on client's monthly salary)
- Able to spend a minimum percentage of the limit allowed (decided by the management) for at least a year with no annual fee on the CC.



Personalized Marketing

- **8 out of 10 customers** prefer businesses that offer personalized experiences (Deloitte, 2023)
- Tailored communication channel as per individual client preferences and needs
- Offering the POS lending options, like Apple's Pay Later, which helps increase transaction volume and customer experience



Relationship Marketing

- Point-based loyalty program, membership subscriptions, incentives
- Collecting customer feedback to improve products and services quality and satisfaction.

Thank You!

Appendices

Table 1.a. Linear Regression Coefficient Table

Note: All Variables in data set.

Model Summary ^b									
Model	R	R Square	Adjusted R Square	Std. Error of the Estimate	R Square Change	Change Statistics			
						F Change	df1	df2	Sig. F Change
1	.757 ^a	.573	.573	.57687	.573	7918.242	8	47241	<.001

a. Predictors: (Constant), Log_Age, Checkings Acc 96, Loan96, Inv Trust 96, Mortgage 96, Credit card 96, Savings 96, Log_Income

b. Dependent Variable: Log_Transactions

Coefficients ^a										
Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.	95.0% Confidence Interval for B		Collinearity Statistics	
		B	Std. Error	Beta			Lower Bound	Upper Bound	Tolerance	VIF
1	(Constant)	.220	.126		1.744	.081	-.027	.468		
	Loan96	.166	.012	.043	13.819	<.001	.142	.189	.950	1.053
	Credit card 96	.163	.008	.062	19.454	<.001	.146	.179	.883	1.132
	Checkings Acc 96	1.037	.124	.025	8.357	<.001	.793	1.280	.984	1.017
	Inv Trust 96	.049	.013	.012	3.876	<.001	.024	.074	.950	1.052
	Mortgage 96	.003	.015	.001	.207	.836	-.027	.033	.962	1.039
	Savings 96	.234	.006	.122	37.060	<.001	.222	.247	.834	1.199
	Log_Income	.396	.002	.679	190.936	<.001	.392	.400	.716	1.397
	Log_Age	-.160	.015	-.034	-10.493	<.001	-.190	-.130	.885	1.130

a. Dependent Variable: Log_Transactions

Appendices

Table 1.b. Linear Regression Coefficient Table

Note: All Significant variables.

Model Summary ^b									
Model	R	R Square	Adjusted R Square	Std. Error of the Estimate	R Square Change	Change Statistics			
						F Change	df1	df2	Sig. F Change
1	.757 ^a	.573	.573	.57687	.573	9049.596	7	47242	<.001

a. Predictors: (Constant), Log_Age, Checkings Acc 96, Loan96, Inv Trust 96, Credit card 96, Savings 96, Log_Income
b. Dependent Variable: Log_Transactions

Coefficients ^a									
Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.	95.0% Confidence Interval for B		Collinearity Statistics
		B	Std. Error	Beta			Lower Bound	Upper Bound	Tolerance VIF
1	(Constant)	.224	.126		1.781	.075	-.022	.469	
	Loan96	.166	.012	.043	13.840	<.001	.142	.189	.952 1.051
	Credit card 96	.163	.008	.062	19.465	<.001	.146	.179	.884 1.131
	Checkings Acc 96	1.033	.123	.025	8.396	<.001	.792	1.275	.999 1.001
	Inv Trust 96	.049	.013	.012	3.886	<.001	.024	.074	.951 1.051
	Savings 96	.234	.006	.122	37.083	<.001	.222	.247	.835 1.198
	Log_Income	.396	.002	.679	191.596	<.001	.392	.400	.721 1.388
	Log_Age	-.160	.015	-.034	-10.491	<.001	-.190	-.130	.886 1.129

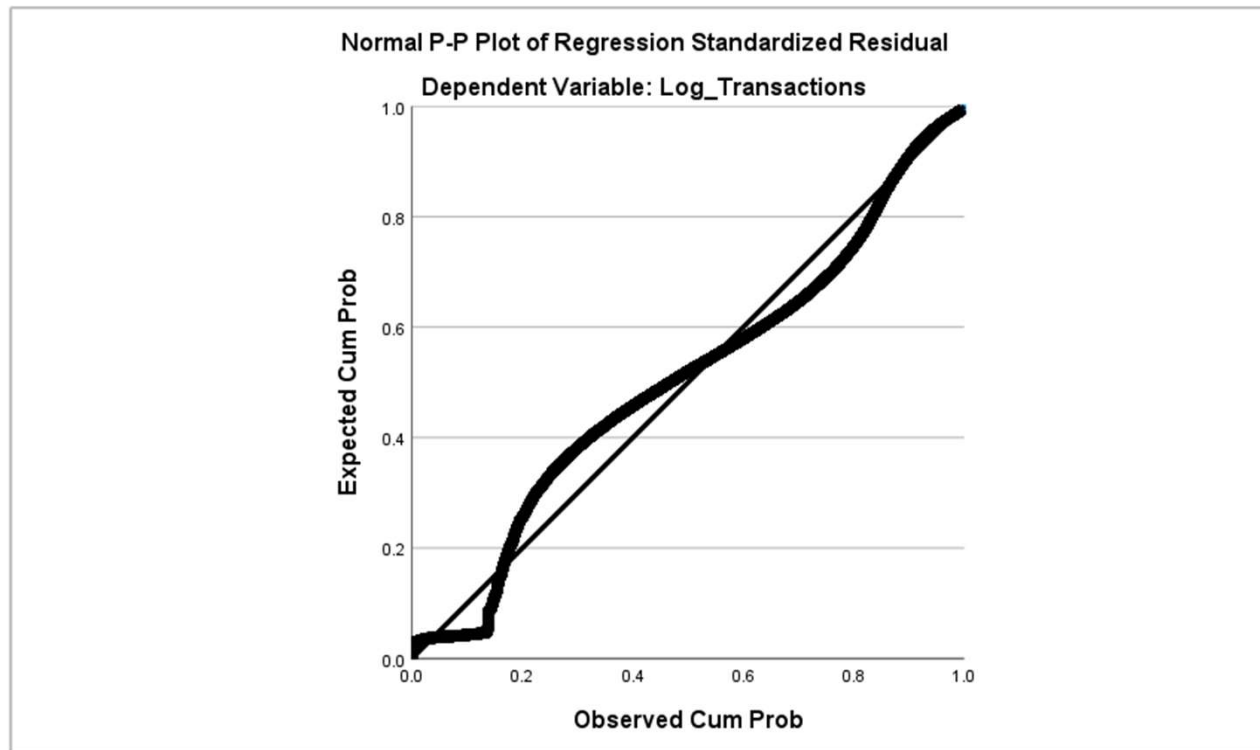
a. Dependent Variable: Log_Transactions

Note: Significant Variables in data set.



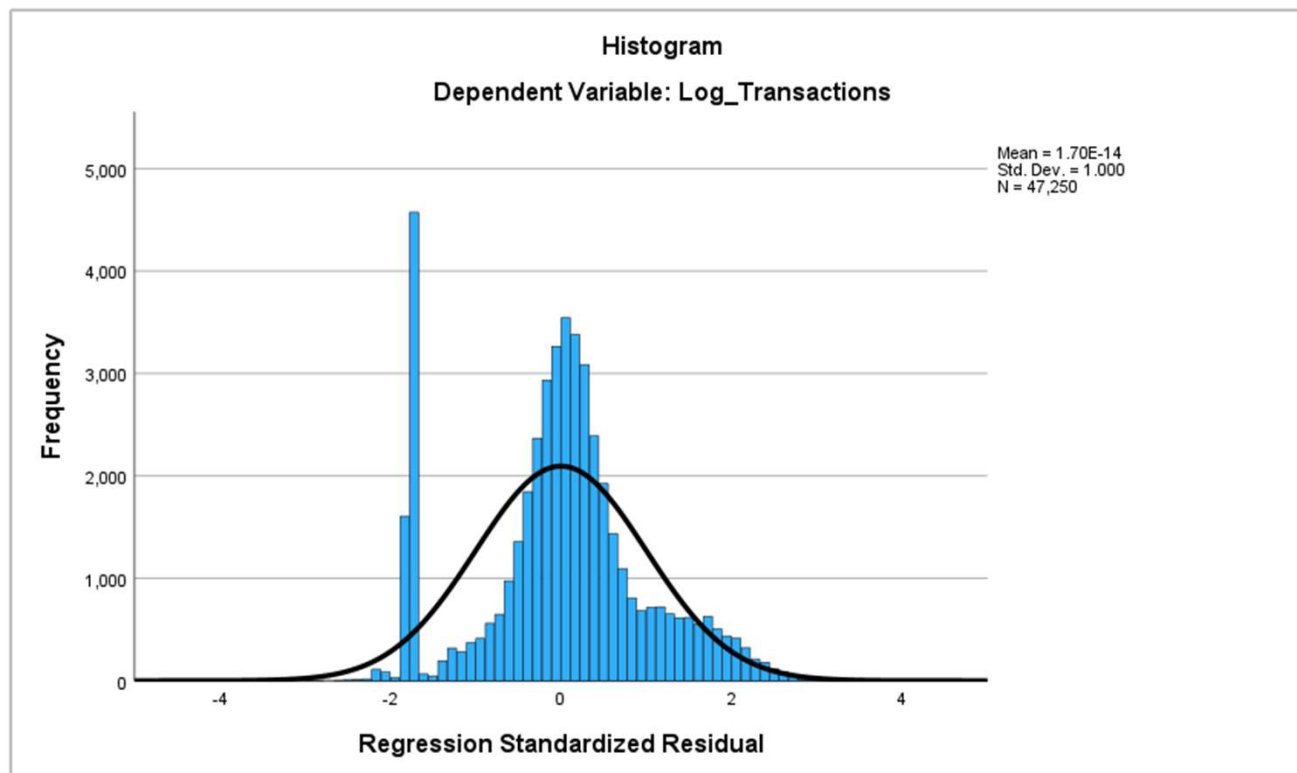
Appendices

Figure 1.a. Normal P-Plot Linear Regression : Activity Levels



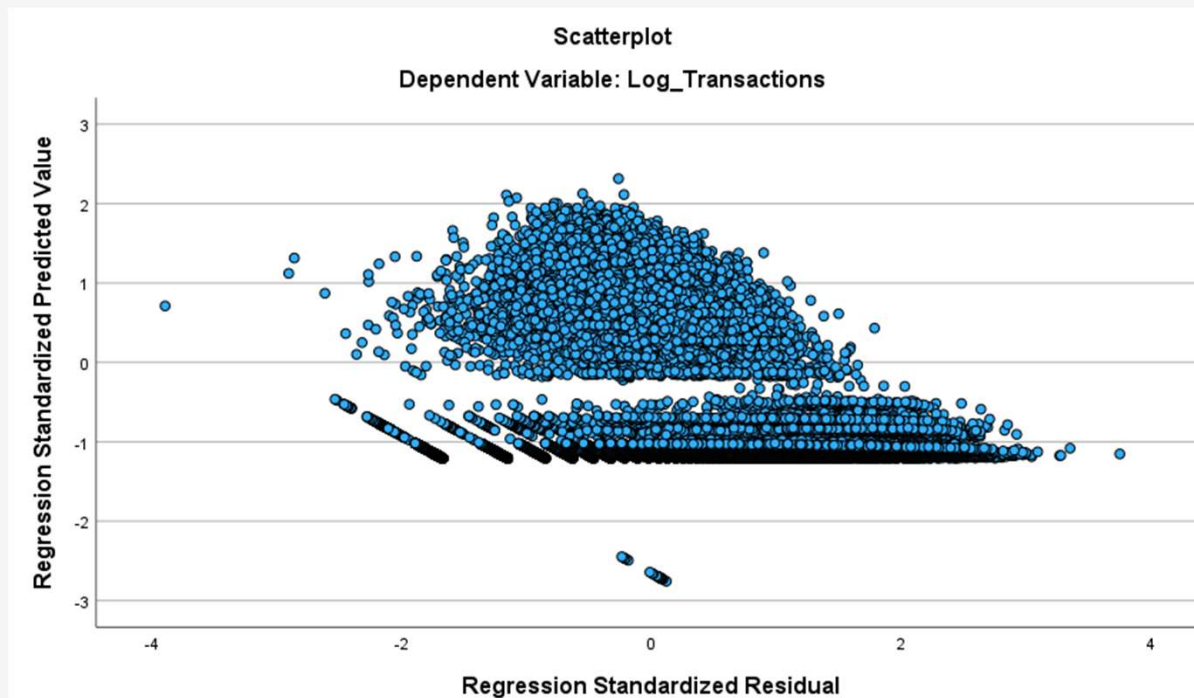
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Figure 1.b. Histogram Linear Regression : Activity Levels



Appendices

Figure 1.c. Scatter Plot Linear Regression: Activity Levels



Appendices

Table 2. T-Test between Churn and LogTransactions:

Hypothesis Tested: There is no significance between number of transactions and customer retention.

Group Statistics					
	churn_t1	N	Mean	Std. Deviation	Std. Error Mean
Log_Transactions	.00	46349	1.7787	.88020	.00409
	1.00	902	1.3684	.91017	.03031

Independent Samples Test											
Levene's Test for Equality of Variances						t-test for Equality of Means					
		F	Sig.	t	df	Significance		Mean Difference	Std. Error Difference	95% Confidence Interval of the Difference	
Log_Transactions	Equal variances assumed	17.106	<.001	13.854	47249	<.001	<.001	.41023	.02961	.35219	.46826
	Equal variances not assumed			13.415	934.090	<.001	<.001	.41023	.03058	.35021	.47024

Independent Samples Effect Sizes					
		Standardizer ^a	Point Estimate	95% Confidence Interval	
Log_Transactions	Cohen's d	.88078	.466	.400	.532
	Hedges' correction	.88079	.466	.400	.532
	Glass's delta	.91017	.451	.382	.520

a. The denominator used in estimating the effect sizes.
Cohen's d uses the pooled standard deviation.
Hedges' correction uses the pooled standard deviation, plus a correction factor.
Glass's delta uses the sample standard deviation of the control group.

Appendices

Table 3. Cross Tabulation – Category Income with Savings

Chi-Square Tests

	Value	df	Asymptotic Significance (2-sided)
Pearson Chi-Square	6543.589 ^a	7	.000
Likelihood Ratio	7042.327	7	.000
Linear-by-Linear Association	6223.247	1	.000
N of Valid Cases	54784		

a. 6 cells (37.5%) have expected count less than 5. The minimum expected count is 1.26.

Cat_Inc * Savings 96 Crosstabulation

		Savings 96		Total
		0	1	
Cat_Inc .00	Count	20205	2940	23145
	% within Cat_Inc	87.3%	12.7%	100.0%
	% within Savings 96	53.8%	17.1%	42.2%
1.00	Count	17207	14087	31294
	% within Cat_Inc	55.0%	45.0%	100.0%
	% within Savings 96	45.8%	81.8%	57.1%
2.00	Count	112	148	260
	% within Cat_Inc	43.1%	56.9%	100.0%
	% within Savings 96	0.3%	0.9%	0.5%
3.00	Count	26	26	52
	% within Cat_Inc	50.0%	50.0%	100.0%
	% within Savings 96	0.1%	0.2%	0.1%
4.00	Count	9	11	20
	% within Cat_Inc	45.0%	55.0%	100.0%
	% within Savings 96	0.0%	0.1%	0.0%
5.00	Count	3	1	4
	% within Cat_Inc	75.0%	25.0%	100.0%
	% within Savings 96	0.0%	0.0%	0.0%
6.00	Count	1	3	4
	% within Cat_Inc	25.0%	75.0%	100.0%
	% within Savings 96	0.0%	0.0%	0.0%
7.00	Count	2	3	5
	% within Cat_Inc	40.0%	60.0%	100.0%
	% within Savings 96	0.0%	0.0%	0.0%
Total	Count	37565	17219	54784
	% within Cat_Inc	68.6%	31.4%	100.0%
	% within Savings 96	100.0%	100.0%	100.0%

Appendices

Table 4. Logistic Regression model summary table highlighting the model fit.

Model Summary			
Step	-2 Log likelihood	Cox & Snell R Square	Nagelkerke R Square
1	8781.115 ^a	.005	.028
2	8755.178 ^a	.005	.031
3	8735.623 ^a	.006	.033
4	8723.903 ^b	.006	.035
5	8717.165 ^b	.006	.036
6	8711.571 ^b	.006	.036
7	8706.511 ^b	.006	.037

a. Estimation terminated at iteration number 7 because parameter estimates changed by less than .001.

b. Estimation terminated at iteration number 8 because parameter estimates changed by less than .001.

Table 5. Omnibus test of logistic regression model coefficients showing the significance of each variable at each step using Chi Square test

Omnibus Tests of Model Coefficients				
		Chi-square	df	Sig.
Step 1	Step	228.428	1	<.001
	Block	228.428	1	<.001
	Model	228.428	1	<.001
Step 2	Step	25.937	1	<.001
	Block	254.365	2	<.001
	Model	254.365	2	<.001
Step 3	Step	19.555	1	<.001
	Block	273.920	3	<.001
	Model	273.920	3	<.001
Step 4	Step	11.719	1	<.001
	Block	285.639	4	<.001
	Model	285.639	4	<.001
Step 5	Step	6.738	1	.009
	Block	292.377	5	<.001
	Model	292.377	5	<.001
Step 6	Step	5.595	1	.018
	Block	297.972	6	<.001
	Model	297.972	6	<.001
Step 7	Step	5.060	1	.024
	Block	303.031	7	<.001
	Model	303.031	7	<.001

Appendices

Table 6. Variables used in our Logistic Regression. B column gives us the weight of the variable and the Exp(B) gives us the percentage change in churn with changes in the variable.

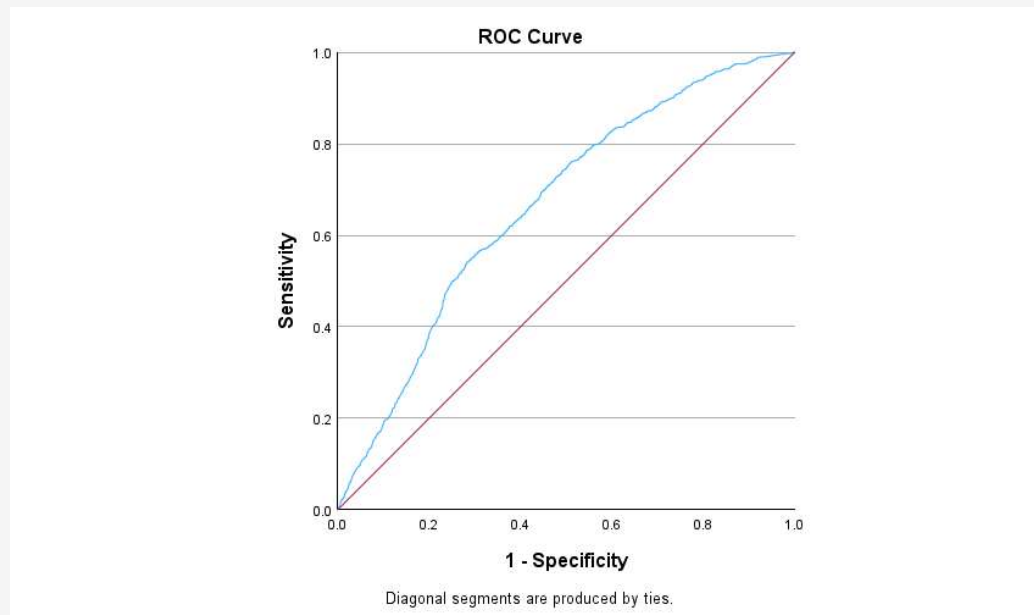
Variables in the Equation									
		B	S.E.	Wald	df	Sig.	Exp(B)	95% C.I. for EXP(B)	
								Lower	Upper
Step 7 ^a	Number transactions 96	-.003	.000	78.732	1	<.001	.997	.996	.998
	Loan96(1)	-.548	.258	4.517	1	.034	.578	.348	.958
	Credit card 96(1)	-.380	.165	5.267	1	.022	.684	.495	.946
	Inv Trust 96(1)	-.847	.294	8.263	1	.004	.429	.241	.764
	Mortgage 96(1)	-.648	.321	4.090	1	.043	.523	.279	.980
	Savings 96(1)	-.469	.101	21.792	1	<.001	.625	.514	.762
	Age 96	.010	.002	19.881	1	<.001	1.010	1.005	1.014
	Constant	-3.759	.092	1675.796	1	<.001	.023		

- a. Variable(s) entered on step 1: Number transactions 96.
- b. Variable(s) entered on step 2: Savings 96.
- c. Variable(s) entered on step 3: Age 96.
- d. Variable(s) entered on step 4: Inv Trust 96.
- e. Variable(s) entered on step 5: Credit card 96.
- f. Variable(s) entered on step 6: Loan96.
- g. Variable(s) entered on step 7: Mortgage 96.

[illegible]

Appendices

Figure 2. ROC analysis for Logistic Regression



Appendices

Figure 3.a. KIWI Bank

Statement of comprehensive income

For the year ended 30 June 2023

Dollars in millions		Note	Year ended 30/06/23	Year ended 30/06/22
Profit after taxation			175	131
Other comprehensive income				
Net gains/(losses) from changes in reserves that may subsequently be reclassified to profit or loss:				
- Fair value reserve (net of tax)	26		15	(66)
- Cash flow hedge reserve (net of tax)	26		(58)	191
Other comprehensive income for the year			(43)	125
Total comprehensive income for the year			132	256

https://media.kiwibank.co.nz/media/documents/General_Disclosure_Statement_Jun23.pdf

Appendices

Figure 3.b. KIWI Bank

Notes to the financial statements continued

6. Operating expenses continued

Dollars in millions	Year ended 30/06/23	Year ended 30/06/22
Salaries and wages	263	227
Other personnel-related costs	48	42
Information technology and system costs	102	94
Premises costs	18	16
Office supplies, postage and telephone	9	10
Consultancy and professional fees	17	16
Marketing and customer communications	20	17
Amortisation and depreciation	41	45
Other expenses	16	13
Total operating expenses	534	480

https://media.kiwibank.co.nz/media/documents/General_Disclosure_Statement_Jun23.pdf

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